# Sikander Mughal

FREELANCER · FULL STACK DEVELOPER



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**Location** Lahore, Pakistan

## Profile

I'm a full stack developer who loves solving software-related challenges, especially the ones where the hard part is working out the right approach rather than just writing the code. I'm open to freelance work and interesting problems, so get in touch and let's get things done.

## Skills

| | |
|---|---|
| LANGUAGES | English, Urdu, Punjabi |
| FRAMEWORKS | Next.js, Flutter, Kotlin, Rust, Capacitor, Playwright |
| TOOLS | Git, Node.js + pnpm, Neovim, PyTorch, OpenCode |

## Projects

### Recorder by Alex

An Android recorder with a live HLS dashboard served by the phone itself, built without a backend.

- Hand-written HTTP server and HLS dashboard served by the phone, with no backend
- Camera frames copied plane by plane into a hardware H.264 encoder, strides included
- Fragmenter injects the missing tfdt and clears the tfhd base offset so MSE will play the stream
- Fixed a black preview that traced back to an empty video track and a rejected moof box
- Foreground service keeps recording with the screen off; 31 unit tests cover the container chain
- About 3 MB signed release APK with R8 and resource shrinking

`Kotlin` `CameraX` `Media3` `H.264`

### Alex Store

A static storefront with no backend: SQLite compiled to WebAssembly in the browser, and semantic search on device.

- SQLite in the browser via vendored sql.js, with no backend and no CDN in the critical path
- 384-dimension semantic search with ternlight in WebAssembly; vectors ship precomputed
- Generated static pages from a zero-dependency build script
- PKR prices stored as integer paisa, with no floating point anywhere
- WhatsApp-only checkout, with the order prefilled from the catalog
- Seven pages, 8 products and 3 collections in about 930 KB with no package manager

`JavaScript` `SQLite` `sql.js` `WebAssembly`

### Alex Store for Android

The Android app for Alex Store, with the same semantic search running natively through Rust and dart:ffi.

- ternlight compiled to native Rust and called through dart:ffi
- Parity test checks 384-dimension vectors and ranking order against the site within 1e-3
- Signed release APK verified by a script that checks the certificate, R8 and model checksum
- Launch video overlay runs under the first Flutter frame so the catalog is ready when it fades
- On-device index built from the same catalog.json the website serves
- No interop codegen: the FFI boundary is a matched pair of Rust and Dart files

`Flutter` `Dart` `Rust` `FFI`

### DoL by Alex

An Android port of an open source Twine interactive fiction game, migrated twice and shipped offline.

- Second full shell migration: Cordova to Capacitor to Flutter 3.47.7
- Fixed export failing silently across four layers at once
- Re-encoded 18,593 PNGs to lossless WebP, cutting artwork 60.9% and the release APK 7.5 MB
- Measured every optimisation instead of assuming: lossy WebP was larger, so it was rejected
- Traced 2,073 silently missing images to percent-encoded asset keys with spaces in paths
- Reuses the previous signing key, so it installs as an in-place update over the old build

`Flutter` `Dart` `Android` `WebView`

### 4 Games by Alex

Four playable games in one Android app, including chess driven by the real Stockfish 19 engine.

- Stockfish 19 lite packaged as a 1.7 MB WASM module inside the APK
- Worker based engine so the interface never blocks while the bot calculates
- Fixed illegal move generation and worker lifecycle bugs found on device
- R8 obfuscation and resource shrinking for release builds

`Kotlin` `Jetpack Compose` `Stockfish` `WebAssembly`

### Photo Editor by Alex

A native Android photo editor built with Compose: layers, filters, text, drawing and cutouts.

- 42 Kotlin files split cleanly into a core engine and a Compose UI layer
- Separate gesture stage per layer type for transform, draw, text and tap select
- Project autosave and a native SAF export sheet
- Background removal through cutout composition

`Kotlin` `Jetpack Compose` `ExifInterface` `Kotlin Coroutines`

## Education

**Intermediate (FA)** 2025

Private student · Lahore, Pakistan

**Matriculation** 2021

CDG Boys High School · Lahore, Pakistan